

Hanjia Lyu

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Research Areas

Computational Social Science, Multimodal Learning, Graph Machine Learning

Education

Ph.D. in Computer Science, University of Rochester	2021–2026
M.S. in Data Science, University of Rochester	2019–2020
B.Econ., Fudan University	2013–2017

Professional Experience

Singapore Management University	<i>Singapore</i>
Assistant Professor, Department of Computing and Information Systems	July 2026–
Snap	<i>Remote</i>
Research Scientist Intern	May 2025–Apr 2026
Meta	<i>Menlo Park, CA</i>
Research Scientist Intern	Jun 2024–Jan 2025 Apr–Aug 2023 May–Sep 2022
Wyze Labs	<i>Remote</i>
Research Scientist Intern	Jan–May 2022
University of Rochester Medical Center	<i>Rochester, NY</i>
Research Assistant	Jan–May 2021

Recognition

◇ Google PhD Fellowship	2024
◇ Donald M. and Janet C. Barnard Fellowship	2023

Conference/Journal Organizing Experience

Area Chair

- ◇ Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) 2025
- ◇ Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024

Senior Program Committee

- ◇ International AAAI Conference on Web and Social Media (ICWSM) 2026–2027

Guest Editor for Journals

- ◇ Frontiers in Public Health
- ◇ ACM Transactions on Multimedia Computing, Communications, and Applications
 - Special Issue on Multimodal Video Understanding and Analysis with Foundation Models

Organizing Committee

- ◇ 1st Multimodal Learning for Social Good (MML4SG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024
- ◇ 1st Visual-Language Alignment in Text-Guided Multi-Modal Generation (TMMG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024

Session Chair

- ◇ International AAAI Conference on Web and Social Media (ICWSM) 2026
- ◇ ACM International Conference on Information and Knowledge Management (CIKM) 2024

Publications

Refereed Journal Papers

1. **Hanjia Lyu***, Jinfa Huang*, Daoan Zhang*, Yongsheng Yu*, Xinyi Mou*, Jinsheng Pan, Zhengyuan Yang, Zhongyu Wei, and Jiebo Luo. GPT-4V(ision) as a social media

analysis engine. *ACM Trans. Intell. Syst. Technol.*, 16(3), April 2025. ISSN 2157-6904. doi: 10.1145/3709005. URL <https://doi.org/10.1145/3709005>

→ Presented at: *Synchronous Reading Club*

2. Weihong Qi, Jinsheng Pan, **Hanjia Lyu**, and Jiebo Luo. Excitements and concerns in the post-ChatGPT era: Deciphering public perception of AI through social media analysis. *Telematics and Informatics*, 2024. URL <https://doi.org/10.1016/j.tele.2024.102158>

→ Media coverage: *The Conversation, Tech Xplore, Futurity*

3. Junyu Chen, Jie An, **Hanjia Lyu**, Christopher Kanan, and Jiebo Luo. Learning to evaluate the artness of AI-generated images. *IEEE Transactions on Multimedia*, 2024. URL <https://doi.org/10.1109/TMM.2024.3410672>
4. **Hanjia Lyu**, Aarsal Imtiaz, Yufei Zhao, and Jiebo Luo. Human behavior in the time of COVID-19: Learning from big data. *Frontiers in Big Data*, 6:1099182, 2023. URL <https://doi.org/10.3389/fdata.2023.1099182>
5. Yangxin Fan, **Hanjia Lyu**, Jin Xiao, and Jiebo Luo. American Twitter users revealed social determinant-related oral health disparities amid the COVID-19 pandemic. *Quintessence International*, 54(1), 2023. URL <https://doi.org/10.3290/j.qi.b3479979>
6. **Hanjia Lyu**, Zihe Zheng, and Jiebo Luo. Misinformation versus facts: Understanding the influence of news regarding COVID-19 vaccines on vaccine uptake. *Health Data Science*, 2022. URL <https://doi.org/10.34133/2022/9858292>

→ Media coverage: *Medical Economics, News Medical, News Wise, Medical Xpress*

→ Presented at: *Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*

7. **Hanjia Lyu**, Yangxin Fan, Ziyu Xiong, Mayya Komisarchik, and Jiebo Luo. Understanding public opinion toward the #StopAsianHate movement and the relation with racially motivated hate crimes in the US. *IEEE Transactions on Computational Social Systems*, 2021. URL <https://doi.org/10.1109/TCSS.2021.3136858>

→ Media coverage: *UofR News Center, Rochester Beacon, Futurity*

8. **Hanjia Lyu**, Junda Wang, Wei Wu, Viet Duong, Xiyang Zhang, Timothy D. Dye, and Jiebo Luo. Social media study of public opinions on potential COVID-19 vaccines: informing dissent, disparities, and dissemination. *Intelligent Medicine*, 2(1):1–12, 2022. ISSN 2667-1026. doi: <https://doi.org/10.1016/j.imed.2021.08.001>. URL <https://www.sciencedirect.com/science/article/pii/S266710262100036X>

→ Media coverage: *News Medical*

→ Presented at: *Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*

9. Ziyu Xiong, Pin Li, **Hanjia Lyu**, and Jiebo Luo. Social media opinions on working from home in the United States during the COVID-19 pandemic: Observational study. *JMIR medical informatics*, 9(7):e29195, 2021. URL <https://doi.org/10.2196/29195>
10. Wei Wu, **Hanjia Lyu**, and Jiebo Luo. Characterizing discourse about COVID-19 vaccines: A Reddit version of the pandemic story. *Health Data Science*, 2021. URL <https://doi.org/10.34133%2F2021%2F9837856>
11. Xupin Zhang, **Hanjia Lyu**, and Jiebo Luo. What contributes to a crowdfunding campaign’s success? evidence and analyses from GoFundMe data. *Journal of Social Computing*, 2(2):183–192, 2021. URL <https://doi.org/10.23919/JSC.2021.0010>

→ Media coverage: *The New York Times, The Wall Street Journal*

12. Xiyang Zhang, Yu Wang, **Hanjia Lyu**, Yipeng Zhang, Yubao Liu, and Jiebo Luo. The influence of COVID-19 on the well-being of people: Big data methods for capturing the well-being of working adults and protective factors nationwide. *Frontiers in Psychology*, 12:681091, 2021. URL <https://doi.org/10.3389%2Ffpsyg.2021.681091>
13. Yipeng Zhang, **Hanjia Lyu**, Yubao Liu, Xiyang Zhang, Yu Wang, and Jiebo Luo. Monitoring depression trends on Twitter during the COVID-19 pandemic: Observational study. *JMIR infodemiology*, 1(1):e26769, 2021. URL <https://doi.org/10.2196/26769>

→ Media coverage: *Tech Xplore, UofR News Center*

14. Siqing Cao, **Hanjia Lyu**, and Xian Xu. Insurtech development: Evidence from Chinese media reports. *Technological Forecasting and Social Change*, 161:120277, 2020. URL <https://doi.org/10.1016/j.techfore.2020.120277>
15. **Hanjia Lyu**, Long Chen, Yu Wang, and Jiebo Luo. Sense and sensibility: Characterizing social media users regarding the use of controversial terms for COVID-19. *IEEE Transactions on Big Data*, 7(6):952–960, 2020. URL <https://doi.org/10.1109%2FTBDATA.2020.2996401>

→ Media coverage: *IEEE Spectrum, Tech Xplore, UofR News Center*

→ Presented at: *Experience Rochester, CSUG Research Panel*

Refereed Conference Papers

1. **Hanjia Lyu**. Building human-aligned AI through representation augmentation and structuring. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, WSDM ’26, New York, NY, USA, 2026. Association for Computing Machinery. ISBN 9798400722929. doi: 10.1145/3773966.3784981. URL <https://doi.org/10.1145/3773966.3784981>

→ Will be presented at: WSDM 2026

2. **Hanjia Lyu**, Jiebo Luo, Jian Kang, and Allison Koenecke. Characterizing bias: Benchmarking large language models in simplified versus traditional Chinese. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '25, pp. 2815–2846, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400714825. doi: 10.1145/3715275.3732182. URL <https://doi.org/10.1145/3715275.3732182>

→ Presented at: FAccT 2025, IC2S2 2025 (Parallel Talk)

3. Chenwei Lin, **Hanjia Lyu**, Xian Xu, and Jiebo Luo. INS-MMBench: A comprehensive benchmark for evaluating lvlms’ performance in insurance. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, Honolulu, HI, USA, October 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2406.09105>
4. Sreejato Chatterjee, Linh Tran, Quoc Duy Nguyen, Roni Kirson, Drue Hamlin, Harvest Aquino, **Hanjia Lyu**, Jiebo Luo, and Timothy Dye. Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models. In *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, December 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2509.15216>
5. Yawen Zheng, **Hanjia Lyu**, and Jiebo Luo. Irony in Emojis: A Comparative Study of Human and LLM Interpretation. In *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, December 2025. IEEE Computer Society. URL <https://arxiv.org/pdf/2501.11241>
6. Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, **Hanjia Lyu**, Joe Barrow, Hongjie Chen, Franck Dernoncourt, et al. From selection to generation: A survey of llm-based active learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://arxiv.org/pdf/2502.11767>
7. Dang Nguyen, Jian Chen, Yu Wang, Gang Wu, Namyong Park, Zhengmian Hu, **Hanjia Lyu**, Junda Wu, Ryan Aponte, Yu Xia, et al. Gui agents: A survey. In *Findings of the Association for Computational Linguistics: ACL 2025*, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://arxiv.org/pdf/2412.13501>
8. Xiaofei Zhou, **Hanjia Lyu**, Yuxin Sa, Mengfan Li, Advait Sarkar, Jiebo Luo, Michael Daley, and Zhen Bai. Empower K-12 teachers to create ml-supported scientific inquiries. In *International Conference on Artificial Intelligence in Education, AIED*, Palermo, Italy, July, 2025

→ Presented at: AIED 2025 (**Best Paper Nominee**)

9. Weihong Qi, **Hanjia Lyu**, and Jiebo Luo. Representation bias in political sample simulations with large language models. In *Companion Proceedings of the ACM on Web*

Conference 2025, WWW '25, pp. 1264–1267, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400713316. doi: 10.1145/3701716.3715591. URL <https://doi.org/10.1145/3701716.3715591>

10. Jinfa Huang*, Jinsheng Pan*, Zhongwei Wan, **Hanjia Lyu**, and Jiebo Luo. Evolver: Chain-of-evolution prompting to boost large multimodal models for hateful meme detection. In *The 31st International Conference on Computational Linguistics*, Abu Dhabi, UAE, 2025. URL <https://arxiv.org/pdf/2407.21004>
11. **Hanjia Lyu**, Song Jiang, Hanqing Zeng, Yinglong Xia, Qifan Wang, Si Zhang, Ren Chen, Chris Leung, Jiajie Tang, and Jiebo Luo. LLM-Rec: Personalized recommendation via prompting large language models. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 583–612, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-naacl.39. URL <https://aclanthology.org/2024.findings-naacl.39/>
 → *Presented at: NAACL 2024, DEFIRST Reading Group at Mila x Vector Institute*
12. Hanqing Zeng*, **Hanjia Lyu***, Diyi Hu, Yinglong Xia, and Jiebo Luo. Mixture of weak & strong experts on graphs. In *The Twelfth International Conference on Learning Representations, ICLR*, Vienna, Austria, May, 2024. URL <https://openreview.net/pdf?id=wYvuY60SdD>
13. **Hanjia Lyu***, Jinsheng Pan*, Zichen Wang*, and Jiebo Luo. Computational assessment of hyperpartisanship in news titles. *Proceedings of the International AAAI Conference on Web and Social Media*, 18(1):999–1012, May 2024. doi: 10.1609/icwsm.v18i1.31368. URL <https://ojs.aaai.org/index.php/ICWSM/article/view/31368>
 → *Media coverage: UofR News Center, Study Finds, ScienceBlog, EurekAlert!*
14. **Hanjia Lyu***, Weihong Qi*, Zhongyu Wei, and Jiebo Luo. Human vs. LMMs: Exploring the discrepancy in emoji interpretation and usage in digital communication. In *AAAI International Conference on Web and Social Media, ICWSM*, Buffalo, NY, USA, June, 2024. URL <https://arxiv.org/pdf/2401.08212.pdf>
15. **Hanjia Lyu**. Towards advancing text-based user and item representation in personalized recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, New York, NY, USA, 2024. URL <https://doi.org/10.1145/3627673.3680270>
16. Zhongyi Qiu, Kangyi Qiu, **Hanjia Lyu**, Wei Xiong, and Jiebo Luo. Semantics Preserving Emoji Recommendation with Large Language Models. In *2024 IEEE International Conference on Big Data (BigData)*, pp. 7131–7140, Los Alamitos, CA, USA, December 2024. IEEE Computer Society. doi: 10.1109/BigData62323.2024.10825696. URL <https://doi.ieeecomputersociety.org/10.1109/BigData62323.2024.10825696>

17. Yiyao Tao*, Hengyu Zhang*, Bali Dey, Selenge Tulga, **Hanjia Lyu**, and Jiebo Luo. In the eyes of the bystander: Are the stances on different conflicts correlated? In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Washington DC, USA, December, 2024. URL <https://arxiv.org/pdf/2401.08733>
18. Junda Wang, Weijian Li, Han Wang, **Hanjia Lyu**, Caroline Thirukumaran, Addisu Mesfin, Hong Yu, and Jiebo Luo. Crtre: Causal rule generation with target trial emulation framework. In *IEEE Big Data Conference*, Washington DC, USA, December, 2024. URL <https://arxiv.org/pdf/2209.14975.pdf>
19. Daoan Zhang, Junming Yang, **Hanjia Lyu**, Zijian Jin, Yuan Yao, Mingkai Chen, and Jiebo Luo. A benchmark and chain-of-thought prompting strategy for large multi-modal models with multiple image inputs. In *International Conference on Pattern Recognition, ICPR*, Kolkata, India, December, 2024. URL <https://arxiv.org/pdf/2401.02582.pdf>
20. Xinnong Zhang, Haoyu Kuang, Xinyi Mou, **Hanjia Lyu**, Kun Wu, Siming Chen, Jiebo Luo, Xuanjing Huang, and Zhongyu Wei. SoMeLVLM: A large vision language model for social media processing. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2366–2389, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.140. URL <https://aclanthology.org/2024.findings-acl.140/>
21. Junyu Chen, Jie An, **Hanjia Lyu**, Christopher Kanan, and Jiebo Luo. Holistic visual-textual sentiment analysis with prior models. In *2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, August, 2024. URL <https://doi.ieeecomputersociety.org/10.1109/MIPR62202.2024.00037>
22. Xiaofei Zhou, Jingwan Tang, **Hanjia Lyu**, Xinyi Liu, Zhenhao Zhang, Lichen Qin, Fiona Au, Advait Sarkar, and Zhen Bai. Creating an authoring tool for K-12 teachers to design ML-supported scientific inquiry learning. In *Late Breaking Papers: ACM Conference on Human Factors in Computing Systems, CHI*, Honolulu, HI, USA, May, 2024. URL <https://doi.org/10.1145/3613905.3650762>
23. Xinyi Mou, Zejun Li, **Hanjia Lyu**, Jiebo Luo, and Zhongyu Wei. Unifying local and global knowledge: Empowering large language models as political experts with knowledge graphs. In *Proceedings of the ACM Web Conference 2024, WWW '24*, pp. 2603–2614, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400701719. doi: 10.1145/3589334.3645616. URL <https://doi.org/10.1145/3589334.3645616>
24. **Hanjia Lyu***, Kevin C. Kapcio*, Kyle C. Purrman, Christian G. Peyre, Carolyn E. Jones, Michal J. Lada, and Jiebo Luo. Applying machine learning to predict esophageal cancer recurrence after esophagectomy. In *IEEE International Conference on Digital Health, ICDH*, Chicago, IL, USA, July, 2023. URL <https://doi.org/10.1109/ICDH60066.2023.00044>

25. Jinsheng Pan, Zichen Wang, Weihong Qi, **Hanjia Lyu**, and Jiebo Luo. Understanding divergent framing of the supreme court controversies: Social media vs. news outlets. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Sorrento, Italy, December, 2023. URL <https://doi.ieeecomputersociety.org/10.1109/BigData59044.2023.10386532>
26. Enting Zhou*, Yurong Liu*, **Hanjia Lyu**, and Jiebo Luo. A fine-grained analysis of public opinion toward Chinese technology companies on Reddit. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Sorrento, Italy, December, 2023. URL <https://doi.ieeecomputersociety.org/10.1109/BigData59044.2023.10386804>
27. Mohammad Mahdi Kamani, Yuhang Yao, **Hanjia Lyu**, Zhongwei Cheng, Lin Chen, Liangju Li, Carlee Joe-Wong, and Jiebo Luo. Wyze rule: Federated rule dataset for rule recommendation benchmarking. In *Thirty-seventh Conference on Neural Information Processing Systems, NeurIPS*, New Orleans, LA, USA, December, 2023. URL <https://openreview.net/pdf?id=qynH28Y4xE>
28. Edinam Kofi Klutse, Samuel Nuamah-Amoabeng, **Hanjia Lyu**, and Jiebo Luo. Dismantling hate: Understanding hate speech trends against nba athletes. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Pittsburgh, PA, USA, September, 2023. URL https://doi.org/10.1007/978-3-031-43129-6_8
29. Jingyang Lin, Junyu Chen, **Hanjia Lyu**, Igor Khodak, Divya Chhabra, Colby L Day Richardson, Irina Prelipcean, Andrew M Dylag, and Jiebo Luo. Predicting adverse neonatal outcomes for preterm neonates with multi-task learning. In *IEEE International Conference on Digital Health, ICDH*, Chicago, IL, USA, July, 2023. URL <https://doi.org/10.1109/ICDH60066.2023.00045>
30. Junyu Chen, Jie An, **Hanjia Lyu**, and Jiebo Luo. How art-like are AI-generated images? an exploratory study. In *Proceedings of the 1st International Workshop on Multimedia Content Generation and Evaluation: New Methods and Practice*, McGE '23, pp. 19–22, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400702785. doi: 10.1145/3607541.3616825. URL <https://doi.org/10.1145/3607541.3616825>
31. Jinsheng Pan*, Weihong Qi*, Zichen Wang, **Hanjia Lyu**, and Jiebo Luo. Bias or diversity? Unraveling fine-grained thematic discrepancy in U.S. news headlines. In *Workshop on News Media and Computational Journalism, MEDIATE, AAAI International Conference on Web and Social Media, ICWSM*, Limassol, Cyprus, June, 2023. URL https://workshop-proceedings.icwsm.org/pdf/2023_25.pdf
 → Media coverage: *UofR News Center, Study Finds, ScienceBlog, EurekAlert!*
32. Xiaofei Zhou, **Hanjia Lyu**, Jiebo Luo, and Zhen Bai. ML-SD modeling: How machine learning can support scientific discovery learning for K-12 stem education. In *Artificial*

Intelligence for Education Workshop AI4Edu, AAAI Conference on Artificial Intelligence, AAAI, Washington DC, USA, February, 2023. URL <https://par.nsf.gov/servlets/purl/10427044>

33. Sajid Hussain Rafi Ahamed, Shahid Shakil, **Hanjia Lyu**, Xinpeng Zhang, and Jiebo Luo. Doctors vs. nurses: Understanding the great divide in vaccine hesitancy among healthcare workers. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Osaka, Japan, December, 2022. URL <https://doi.ieeecomputersociety.org/10.1109/BigData55660.2022.10020853>
34. **Hanjia Lyu** and Jiebo Luo. Understanding political polarization via jointly modeling users, connections and multimodal contents on heterogeneous graphs. In *Proceedings of the 30th ACM International Conference on Multimedia*, MM '22, pp. 4072–4082, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450392037. doi: 10.1145/3503161.3547898. URL <https://doi.org/10.1145/3503161.3547898>
 → *Presented at: MM 2022, Meta, Western New York Image and Signal Processing Workshop*
35. Wei Zhu, Ziheng Zheng, Haitian Zheng, **Hanjia Lyu**, and Jiebo Luo. Learning to aggregate and refine noisy labels for visual sentiment analysis. In *International Conference on Pattern Recognition, ICPR*, Montréal, Canada, August, 2022. URL <https://doi.ieeecomputersociety.org/10.1109/ICPR56361.2022.9956328>
36. Xiaofei Zhou, Jingwan Tang, Beilei Guo, **Hanjia Lyu**, and Jiebo Luo. Challenges and design opportunities in data analysis for ml-empowered scientific inquiry – insights from a teacher professional development study. In *International Conference of the Learning Sciences, ICLS*, Virtual, June, 2022. URL <https://repository.isls.org/handle/1/9077>
37. Aarsal Imtiaz, Danish Khan, **Hanjia Lyu**, and Jiebo Luo. Taking sides: Public opinion over the Israel-Palestine conflict in 2021. In *International Workshop on Social Sensing, SocialSens: Special Edition on Belief Dynamics, AAAI International Conference on Web and Social Media, ICWSM*, Atlanta, GA, USA and online, June, 2022. URL https://workshop-proceedings.icwsml.org/pdf/2022_30.pdf
38. Yichi Qian, Qiyi Shan, **Hanjia Lyu**, and Jiebo Luo. Look behind the censorship: Reposting-user characterization and muted-topic restoration. In *International Workshop on Social Sensing, SocialSens: Special Edition on Belief Dynamics, AAAI International Conference on Web and Social Media, ICWSM*, Atlanta, GA, USA and online, June, 2022. URL https://workshop-proceedings.icwsml.org/pdf/2022_29.pdf
39. Tanqiu Jiang, Sidhant Bendre, **Hanjia Lyu**, and Jiebo Luo. From static to dynamic prediction: Wildfire risk assessment based on multiple environmental factors. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Virtual, December, 2021. URL <https://doi.org/10.1109/BigData52589.2021.9672044>

40. Xupin Zhang, **Hanjia Lyu**, and Jiebo Luo. Understanding the hoarding behaviors during the COVID-19 pandemic using large scale social media data. In *Special Session on Intelligent Data Mining, IEEE Big Data Conference*, Virtual, December, 2021. URL <https://doi.ieeecomputersociety.org/10.1109/BigData52589.2021.9671533>
 → Media coverage: *Tech Xplore*, *UofR News Center*
41. Long Chen, **Hanjia Lyu**, Tongyu Yang, Yu Wang, and Jiebo Luo. Fine-grained analysis of the use of neutral and controversial terms for COVID-19 on social media. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Virtual, July, 2021. URL https://doi.org/10.1007/978-3-030-80387-2_6
 → Media coverage: **IEEE Spectrum**, *Tech Xplore*, *UofR News Center*
42. Karan Vombatkere, **Hanjia Lyu**, and Jiebo Luo. How political is the spread of COVID-19 in the United States? An analysis using transportation and weather data. In *International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation, SBP-BRiMS*, Virtual, July, 2021. URL https://doi.org/10.1007/978-3-030-80387-2_2

Abstracts & Conference Presentations

1. Shazia Siddiqi, Tiffany Panko, Wyatt Hall, Lisette Alcantara, **Hanjia Lyu**, Jiebo Luo, Eva Pressman, and Timothy Dye. Multi-gendered, global analysis of reproductive intention among deaf, hard of hearing, and hearing populations. In *American Public Health Association (APHA) Annual Meeting and Expo*, Washington, D.C., 2025. URL <https://apha.confex.com/apha/2025/meetingapp.cgi/Paper/576909>. Conference abstract
2. Timothy Dye, Brooke Levandowski, Monica Barbosu, Lisette Alcantara, Esteban Avendaño, Jiebo Luo, **Hanjia Lyu**, Saloni Sharma, Nazifa Shinwarie, Shazia Siddiqi, Ivelisse Rivera, and Eva Pressman. Potential covid-19 impacts of lived experience on ecosystems of parenting families worldwide: Results of a multilingual global survey from 168 countries. In *American Public Health Association (APHA) Annual Meeting and Expo*, Washington, D.C., 2025. URL <https://apha.confex.com/apha/2025/meetingapp.cgi/Paper/578291>. Conference abstract
3. Timothe Dye, Lisette Alcántara, **Hanjia Lyu**, Shazia Siddiqi, Saloni Sharma, Eva Pressman, and Jiebo Luo. Oppression, COVID vaccination, and vaccine sentiments in a global sample. *Annals of Epidemiology*, 75:82, 2022. URL <https://www.sciencedirect.com/science/article/am/pii/S1047279722001983>. Conference abstract

Talks & Publicity

◇ Presentation at the ACM International Conference on Web Search and Data Mining (WSDM)	Feb 2026
◇ Presentation at the ACM Conference on Fairness, Accountability, and Transparency (FAccT)	Jun 2025
◇ Invited talk at Meta	Jun 2025
◇ Presentation at the ACM International Conference on Information and Knowledge Management (CIKM)	Oct 2024
◇ Presentation at the IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)	Aug 2024
◇ Presentation at the International AAAI Conference on Web and Social Media (ICWSM)	Jun 2024
◇ Presentation at the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)	Jun 2024
◇ Presentation at the International Conference on Learning Representations (ICLR)	May 2024
◇ Invited talk at Data Science Snack Series: Internship Panel, University of Rochester	Feb 2024
◇ Invited talk at Synchronous Reading Club	Jan 2024
◇ Invited talk at Fudan University	Nov 2023
◇ Invited talk at DEFIRST Reading Group at Mila x Vector Institute	Oct 2023
◇ Presentation at the IEEE International Conference on Digital Health (ICDH)	Jul 2023
◇ Presentation at the AS&E Graduate Research Symposium, University of Rochester	Mar 2023
◇ Invited talk at the Western New York Image and Signal Processing Workshop	Nov 2022
◇ Invited talk at Meta	Aug 2022
◇ Invited talk at the Great Lakes Science Boot Camp for Librarians	Jun 2022
◇ Invited talk at Fudan-UC Center on Contemporary China, UCSD	May 2022
◇ Invited talk at InsurTech Innovation: UK & China as Hotspots of the Global World	Apr 2022
◇ Invited talk at CSUG Research Panel, University of Rochester	Oct 2021
◇ Presentation at the International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS)	Jul 2021
◇ Invited talk at Experience Rochester, University of Rochester	Sep 2020